Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/02/2016
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Global Transcriptome Profiling to Identify Cellular Stress Mechanisms Responsible for Spaceflight-Induced Antibiotic Resistance		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	10/01/2014	**End Date:**	09/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Saylor, Russell	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fajardo-Cavazos, Patricia Ph.D. (University of Florida)		
Grant/Contract No.:	NNX14AT38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Responses of various microorganisms to the spaceflight environment have been examined by numerous studies conducted either in space or on the ground in clinostats. The results clearly show that several microbial behaviors differ in space, but not in a consistent way. The underlying mechanism(s) involved are not explainable by a single model. Exposure to the spaceflight environment (e.g., microgravity, radiation) alters a microbial cell's immediate surroundings, lowering convective mass and heat transfer or reducing mechanical shear forces. Changes in such fundamental physical forces affect the rates at which gases, nutrients, signaling molecules, and waste products are exchanged between microbes and their surroundings. Microbes perceive these alterations as environmental stress (i.e., the "spaceflight syndrome") and mount a complex set of stress responses. One consequence of the spaceflight stress response is alteration in their susceptibility to antibiotics. This could be due to mechanisms such as: production of biofilms, alteration of cell surfaces, up-regulation of drug efflux systems, or an increased rate of mutation in genes encoding drug targets. All of these mechanisms have been well documented as outputs of global microbial stress responses. Of particular interest is oxidative stress, which has been shown both to trigger maladaptive responses in higher organisms including astronauts, and to increase bacterial resistance to antibiotics. Central Hypothesis: Microorganisms subjected to the integrated spaceflight environment invoke a spectrum of stress responses, some leading to alterations in their antibiotic susceptibility. The underlying mechanisms can be identified using transcriptome profiling of model organisms. Specific Aims: We postulate that spaceflight stress responses vary depending on the organism studied because each species of microbe is endowed with its own unique collection of stress response systems. In order to probe the underlying mechanism(s) involved it will be necessary to study spaceflight stress response at the genomic level using model organisms. Specifically we plan to: A. Expose the model bacteria Bacillus subtilis and Staphylococcus epidermidis to: (i) simulated microgravity in the laboratory and (ii) actual spaceflight in 2 parallel Biological Research in Canisters (BRIC) canisters (BRIC-A and BRIC-B) aboard the International Space Station (ISS), with appropriate ground controls. Cells cultivated in BRIC-A will be fixed rapidly with RNA Later and cells in BRIC-B will be frozen in a viable state for return to Earth. B. Total RNA will be extracted from cells in BRIC-A and subjected to transcriptome analysis using Whole Transcriptome Shotgun Sequencing (a.k.a. RNA sequencing or "RNA-Seq") at the University of Florida Interdisciplinary Center for Biotechnology Research (UF-ICBR), to identify the suite of stress responses induced by exposure to spaceflight or simulated microgravity. Cells in BRIC-B will be thawed and immediately assayed for (i) viability, (ii) resistance levels to a battery of antibiotics in the Omnilog system, and (iii) rates of spontaneous mutation to antibiotic resistance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Evidence is mounting that exposure of microorganisms to spaceflight provokes a variety of stress responses, some of which can lead to increased antibiotic resistance. The central goal of this proposal is to identify these stress responses, using high-throughput global transcriptome and phenotypic analyses. Understanding how potentially pathogenic microbes develop resistance to antibiotics in response to stress has significance both for long-term human space exploration and for medical science on Earth.
Task Progress:	Introduction: This report documents the second year activities related to the spaceflight experiment informally referred to here as “Biological Research in Canisters #21 (BRIC-21), with the aim of understanding how bacterial cells perceive and respond to the human spaceflight environment, particularly regarding their development of antibiotic resistance. Post-experiment sample processing: Samples were processed for (i) viable counts; (ii) isolation of rifampicin-resistant (RFMR) mutants; (iii) total RNA isolation; or (iv) Omnilog phenotype microarray profiling. (i) Viable counts. B. subtilis cultures were diluted serially tenfold in PBS buffer, dilutions plated on TSY agar plates, and colonies counted after incubation at 37°C for 24 hours. Viable counts were calculated for Flight (FL) and Ground Control (GC) samples. Because samples were unpaired, non-parametric statistics were applied (Kruskal-Wallis). Conclusion: cells grew to essentially the same concentration in FL and GC samples. (ii) Rifampicin-resistant (RFMR) mutants: The frequency of mutation to RFMR was calculated by dividing the total number of RFMR mutants by the total number of viable cells from each of 3 cultures. Conclusion: FL cultures demonstrated a significantly higher mutation frequency to RFMR than did GC cultures. Individual RFMR mutants have been streak-purified and saved as frozen -70°C glycerol stocks for further characterization by DNA sequencing. We plan to thaw and process additional PDFUs from this experiment to obtain higher numbers of mutants for sequencing analyses. (iii) RNA–seq. B. subtilis cells were thawed on ice, recovered by centrifugation at 0°C, and immediately processed on ice for total RNA extraction. Three entire cultures from FL and GC were processed. Total RNA was isolated and sent to Hudson Alpha Institute to perform RNA-seq. Hudson Alpha performed QC on the samples and provided the following yield and RNA Integrity Numbers (RIN). Conclusion: Sufficient high-quality RNA was isolated to perform RNA-seq. RNA-seq was performed at Hudson Alpha Institute. Graduate student Michael Morrison has been tasked with analysis of the data. Before meaningful analyses could be performed, Michael developed and used the following pipeline to process the raw data: 1. Raw data were downloaded from the Hudson Alpha server using the software Cygwin. 2. Raw data, consisting of 12 raw data files (2 for each sample, corresponding to the Forward and Reverse strands), were uploaded to the UF Galaxy server. 3. Each file was run through FastQ Groomer, to ensure that the sequences were all in the correct format for further processing. 4. Each file was run through FastQ Trimmer, which removed random sequences from the first 12 bases of the 5’ end of each sequence. 5. Files were run through FastQC; Read QC to identify low-quality sequences. 6. If low-quality sequences were detected, the file was run through Steps 4 and 5 again.

	7. Paired files were run through Bowtie2, which mapped the sequences to the most- recently annotated B. subtilis genome sequence. 8. The Bowtie2 results were run through the program “sam-stats” for quality control on the alignments. 9. The Bowtie2 files were then run through the program “htseq”, which counted the number of alignments which mapped to each gene. 10. All of the ‘htseq’ count files were compiled into a master Excel spreadsheet for further comparison and analysis. Conclusion: analysis of processed data is currently in progress. (iv) Omnilog phenotype microarrays: frozen samples were thawed, washed once in Phosphate-Buffered Saline, diluted to the appropriate cell concentration in Inoculating Fluid, and inoculated into PM-11, PM-12, and PM-13 plates containing various concentrations of 72 different antibiotics. Plates were incubated at 37°C in the Omnilog instrument (Biolog, Inc.) and plate images recorded at 15-min intervals for 24 hours. Antibiotic resistance profiles were compared between FL and GC samples for significant differences. FL and GC samples showed no difference in susceptibility to nearly all of the antibiotics tested, with the exceptions of enoxacin and nalidixic acid (NOTE: 6-mercaptopurine is an immunosuppressant anti-cancer drug, not an antibiotic). It is interesting to note that both enoxacin and nalidixic acid belong to the class of fluoroquinolone antibiotics, which act by inhibiting DNA gyrase. Closer inspection of the resistance patterns to these two antibiotics in FL vs. GC samples confirmed that FL samples were more resistant than GC samples.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
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